

Reinforcement Learning work

Introduction

You have the code from the laboratory on the virtual campus. You have 3 files:

- Deep_Reinforcement_Learning.ipynb
- network.py
- dqn_agent.py

The network.py file is the definition of the neural network we will use for deep reinforcement learning. dqn_agent.py is a simple implementation of a Deep Q-Learning agent. The notebook, obviously, shows how it all works together.

Your work

As you can see, the implementation solves the LunarLander scenario of OpenAI's Gym (<https://www.gymnasium.dev/>). Your goal is to modify it, to solve another of Box2D scenarios: CarRacing (https://gymnasium.farama.org/environments/box2d/car_racing/)

To achieve it you'll have to modify the code that receives the state from the environment to suit CarRacing different state structure, as well as the possible actions. And, also, the structure of the neural network (don't need to touch network.py, as its all parameter controlled), but you'll have to adapt the creation of the network.